

Understanding Ownership

```
ubuntu@ip-172-31-26-223:~$ cd
ubuntu@ip-172-31-26-223:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-26-223:~$ ls -l
total 12
-r----- 1 ubuntu ubuntu  0 Feb 18 13:34 devops.txt
-rw-r----- 1 ubuntu ubuntu 12 Feb 18 13:35 notes.txt
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 18 14:06 project
-r----- 1 ubuntu ubuntu 20 Feb 18 13:38 script.sh
ubuntu@ip-172-31-26-223:~$ |
```

Owner: ubuntu

Group: ubuntu

Ubuntu is the owner of the files.

Owner is a specific user who created the file whereas group is a set of users who has the same access right which is assigned to the group.

Basic chown Operations

```
ubuntu@ip-172-31-26-223:~$ touch devops-file.txt
ubuntu@ip-172-31-26-223:~$ ls -l devops-file.txt
-rw-rw-r-- 1 ubuntu ubuntu 0 Feb 18 14:37 devops-file.txt
ubuntu@ip-172-31-26-223:~$ |
```

The owner of devops-file.txt is **ubuntu**

```
ubuntu@ip-172-31-26-223:~$ ls -l devops-file.txt
-rw-rw-r-- 1 ubuntu ubuntu 0 Feb 18 14:37 devops-file.txt
ubuntu@ip-172-31-26-223:~$ chown tokyo devops-file.txt
chown: changing ownership of 'devops-file.txt': Operation not permitted
ubuntu@ip-172-31-26-223:~$ sudo chown tokyo devops-file.txt
ubuntu@ip-172-31-26-223:~$ ls -l devops-file.txt
-rw-rw-r-- 1 tokyo ubuntu 0 Feb 18 14:37 devops-file.txt
ubuntu@ip-172-31-26-223:~$ sudo chown berlin devops-file.txt
ubuntu@ip-172-31-26-223:~$ ls -l devops-file.txt
-rw-rw-r-- 1 berlin ubuntu 0 Feb 18 14:37 devops-file.txt
ubuntu@ip-172-31-26-223:~$ |
```

Basic chgrp Operations

```
ubuntu@ip-172-31-26-223:~$ touch team-notes.txt
ubuntu@ip-172-31-26-223:~$ ls -l team-notes.txt
-rw-rw-r-- 1 ubuntu ubuntu 0 Feb 18 14:43 team-notes.txt
ubuntu@ip-172-31-26-223:~$ sudo groupadd heist-team
ubuntu@ip-172-31-26-223:~$ sudo chgrp heist-team team-notes.txt
ubuntu@ip-172-31-26-223:~$ ls -l team-notes.txt
-rw-rw-r-- 1 ubuntu heist-team 0 Feb 18 14:43 team-notes.txt
ubuntu@ip-172-31-26-223:~$
```

Combined Owner & Group Change

```
ubuntu@ip-172-31-26-223:~$ touch project-config.yaml
ubuntu@ip-172-31-26-223:~$ ls -l project-config.yaml
-rw-rw-r-- 1 ubuntu ubuntu 0 Feb 18 14:46 project-config.yaml
ubuntu@ip-172-31-26-223:~$ sudo chown professor:heist-team project-config.yaml
ubuntu@ip-172-31-26-223:~$ ls -l
total 12
-rw-rw-r-- 1 berlin ubuntu 0 Feb 18 14:37 devops-file.txt
-r----- 1 ubuntu ubuntu 0 Feb 18 13:34 devops.txt
-rw-r----- 1 ubuntu ubuntu 12 Feb 18 13:35 notes.txt
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 18 14:06 project
-rw-rw-r-- 1 professor heist-team 0 Feb 18 14:46 project-config.yaml
-r----- 1 ubuntu ubuntu 20 Feb 18 13:38 script.sh
-rw-rw-r-- 1 ubuntu heist-team 0 Feb 18 14:43 team-notes.txt
ubuntu@ip-172-31-26-223:~$ ls -l project-config.yaml
-rw-rw-r-- 1 professor heist-team 0 Feb 18 14:46 project-config.yaml
ubuntu@ip-172-31-26-223:~$
```

Combined Owner & Group Change

```
ubuntu@ip-172-31-26-223:~$ mkdir app-logs
ubuntu@ip-172-31-26-223:~$ ls -l app-logs/
total 0
ubuntu@ip-172-31-26-223:~$ ls -ld app-logs/
drwxrwxr-x 2 ubuntu ubuntu 4096 Feb 18 14:49 app-logs/
ubuntu@ip-172-31-26-223:~$ sudo chown berlin:heist-team app-logs/
ubuntu@ip-172-31-26-223:~$ ls -l app-logs/
total 0
ubuntu@ip-172-31-26-223:~$ ls -ld app-logs/
drwxrwxr-x 2 berlin heist-team 4096 Feb 18 14:49 app-logs/
ubuntu@ip-172-31-26-223:~$
```

Recursive Ownership

```
ubuntu@ip-172-31-26-223:~$ mkdir -p heist-project/vault
ubuntu@ip-172-31-26-223:~$ mkdir -p heist-project/plans
ubuntu@ip-172-31-26-223:~$ touch heist-project/vault/gold.txt
ubuntu@ip-172-31-26-223:~$ touch heist-project/plans/strategy.conf
ubuntu@ip-172-31-26-223:~$ sudo groupadd planners
ubuntu@ip-172-31-26-223:~$ ls -lR heist-project/
heist-project/:
total 8
drwxrwxr-x 2 ubuntu ubuntu 4096 Feb 18 14:54 plans
drwxrwxr-x 2 ubuntu ubuntu 4096 Feb 18 14:53 vault

heist-project/plans:
total 0
-rw-rw-r-- 1 ubuntu ubuntu 0 Feb 18 14:54 strategy.conf

heist-project/vault:
total 0
-rw-rw-r-- 1 ubuntu ubuntu 0 Feb 18 14:53 gold.txt
ubuntu@ip-172-31-26-223:~$ sudo chown -R professor:planners heist-project/
ubuntu@ip-172-31-26-223:~$ ls -lR heist-project/
heist-project/:
total 8
drwxrwxr-x 2 professor planners 4096 Feb 18 14:54 plans
drwxrwxr-x 2 professor planners 4096 Feb 18 14:53 vault

heist-project/plans:
total 0
-rw-rw-r-- 1 professor planners 0 Feb 18 14:54 strategy.conf

heist-project/vault:
total 0
-rw-rw-r-- 1 professor planners 0 Feb 18 14:53 gold.txt
ubuntu@ip-172-31-26-223:~$
```

Practice Challenge

```
ubuntu@ip-172-31-26-223:~$ sudo groupadd vault-team
ubuntu@ip-172-31-26-223:~$ sudo groupadd tech-team
ubuntu@ip-172-31-26-223:~$ mkdir bank-heist/
ubuntu@ip-172-31-26-223:~$ touch bank-heist/access-codes.txt
ubuntu@ip-172-31-26-223:~$ touch bank-heist/blueprints.pdf
ubuntu@ip-172-31-26-223:~$ touch bank-heist/escape-plan.txt
ubuntu@ip-172-31-26-223:~$ sudo chown tokyo:vault-team bank-heist/access-codes.txt
ubuntu@ip-172-31-26-223:~$ sudo chown berlin:tech-team bank-heist/blueprints.pdf
ubuntu@ip-172-31-26-223:~$ sudo chown nairobi:vault-team bank-heist/escape-plan.txt
ubuntu@ip-172-31-26-223:~$ ls -l bank-heist/
total 0
-rw-rw-r-- 1 tokyo vault-team 0 Feb 18 15:01 access-codes.txt
-rw-rw-r-- 1 berlin tech-team 0 Feb 18 15:01 blueprints.pdf
-rw-rw-r-- 1 nairobi vault-team 0 Feb 18 15:02 escape-plan.txt
ubuntu@ip-172-31-26-223:~$
```

